# README

## Project Information

- **Student**: QUYETDT3
- **Project**: Physical Design Implementation of a 32-Bit CPU
- **Technology**: 45nm CMOS
- **Design Top**: `cpu_32bit_top`

## Delivery Package

The final delivery package is organized as follows:

```
<student_name>_final_delivery/
    - 01_db_checkpoints/
    - 02_export_data/
    - 03_reports/
    - 04_images/
    - 05_scripts/
    - 06_source_files/
    - 07_readme/
    - 08_presentation/
    - 09_scripts/
```

## Folder Description

### 01_db_checkpoints

This folder contains the main Innovus database checkpoints for each implementation stage:

- `floorplan.<your-name>.cui`
- `placement.<your-name>.cui`
- `cts.<your-name>.cui`
- `route.<your-name>.cui`
- `final.<your-name>.cui`

### 02_export_data

This folder contains the final export data of the design:

- `cpu32_final.<your-name>.gds`
- `cpu32_final.<your-name>.spef`
- `cpu32_final.<your-name>.v`
- `cpu32_final.<your-name>.sdf`
- `cpu32_final.<your-name>.def`
- `cpu32_final.<your-name>.lef`

### 03_reports

This folder contains the main reports used for design analysis and signoff review:

- `floorplan_summary.<your-name>.rpt`
- `placement_timing.<your-name>.rpt`
- `placement_density.<your-name>.rpt`
- `cts_clock_summary.<your-name>.rpt`
- `cts_timing.<your-name>.rpt`
- `route_timing.<your-name>.rpt`
- `route_drc.<your-name>.rpt`
- `route_congestion.<your-name>.rpt`
- `final_timing_setup.<your-name>.rpt`
- `final_timing_hold.<your-name>.rpt`
- `final_area.<your-name>.rpt`
- `final_qor_summary.<your-name>.rpt`

### 04_images

This folder contains screenshots of important physical design stages:

- `floorplan.<your-name>.png`
- `powerplan.<your-name>.png`
- `placement.<your-name>.png`
- `congestion_map.<your-name>.png`
- `cts_clock_tree.<your-name>.png`
- `routed_layout.<your-name>.png`
- `timing_path_example.<your-name>.png`
- `drc_marker_example.<your-name>.png`
- `final_chip.<your-name>.png`

### 05_scripts

This folder contains additional support scripts if needed.

### 06_source_files

This folder contains the main source files used for the project, such as:

- netlist
- SDC
- LEF / technology files
- any required setup files

### 07_readme

This folder contains this README file.

### 08_presentation

This folder contains the final presentation file.

### 09_scripts

This folder contains the main implementation scripts:

- `floorplan.tcl`
- `placement.tcl`
- `cts.tcl`
- `global_route.tcl`
- `detail_route.tcl` or `route.tcl`
- `export.tcl`

## Design Flow Summary

The physical design flow was completed in the following order: 1. Floorplan 2. Power planning 3. Placement 4. Clock Tree Synthesis (CTS) 5. Global routing 6. Detail routing 7. Post-route timing and DRC repair 8. Final export

## Notes

- All reports, images, and exported data are grouped by stage for easier review.
- The root clock buffer was preserved and placed near the central sequential region to support balanced clock distribution.
- The final package includes both implementation scripts and final generated outputs.
- File names should be updated with the student name before final submission.

## Submission Reminder

Before submission, please verify:

- all required folders are present
- all required reports are included
- all screenshots are included
- all scripts are included
- final export data is included
- PowerPoint presentation is included